

Statistical Analysis Report: Physical Uncertainty in Logistics World Models

Overview

This report presents a statistical analysis of product dimensions within the e-commerce sector, utilizing the **Amazon ML Challenge 2023** dataset. The main objective is to quantify the physical uncertainty associated with various product categories. Understanding these statistical distributions is a prerequisite for building robust "World Models" in warehouse automation, where robotic agents must estimate physical parameters like size and weight from visual metadata to bridge the "Sim-to-Real" gap.

Dataset Description

The dataset represents a diverse catalog of over 340,000 unique Amazon products. It consists of six columns, including categorical identifiers (PRODUCT_TYPE_ID) and numerical physical measurements (PRODUCT_LENGTH). For this analysis, a filtered subset of 500+ rows was used to ensure statistical relevance. The primary focus was examining how the "length" of a product varies across different types, which directly impacts robotic grasping and storage optimization strategies.

Methods

Following the framework of **Initial Data Analysis (IDA)** proposed by *Bauman et al. (2016)*, the data underwent a cleaning process to address extreme outliers (e.g., erroneous lengths exceeding planetary scales) caused by unit inconsistencies.

Descriptive statistics were used to identify central tendencies, revealing a high degree of skewness. Three visualizations—a log-scaled boxplot, a histogram, and a violin plot—were generated to observe the distribution shapes. Finally, **Levene's test** was selected for hypothesis testing. This test is particularly appropriate for this dataset because it is robust against non-normal distributions (*Levene, 1960*), allowing us to evaluate whether the variance

(uncertainty) is consistent across different product groups.

Results

The analysis revealed a profound non-uniformity in physical dimensions. The Levene's test yielded a statistic of **64.7690** with an extremely low p-value of **8.7635e-16**.

Hypothesis Test Results:

- **Null Hypothesis (H_0):** Variance of product length is equal across categories.
- **Test Statistic (F):** 64.7690
- **P-value:** 8.7635e-16
- **Decision:** Reject H_0 (Significant difference in variance exists).

Visual evidence from the boxplots confirmed that while some categories (e.g., standardized office supplies) show very low variance, others exhibit wide spreads and numerous outliers, indicating high inherent uncertainty.

Interpretation for a Non-Technical Audience

In simple terms, our study proves that "one size does not fit all" in warehouse automation. I found that some types of items are very predictable in their size, while others are highly unpredictable. For a robot, this means it cannot use the same level of confidence for every item it picks up. The near-zero p-value confirms that this isn't a random coincidence but a fundamental characteristic of the inventory. To improve efficiency, robots should be programmed to be more "cautious" and allow for more error margin when handling items from categories I have identified as high-variance.

Limitations and Potential Bias

A primary limitation is the lack of explicit unit labels (mm, cm, inches) in the raw data, which necessitated the removal of extreme values during IDA. This might have excluded valid but unusual oversized items. Furthermore, the dataset is biased toward consumer goods sold on Amazon; therefore, the specific variance levels identified may not directly apply to heavy industrial logistics or specialized medical supply chains without further domain-specific calibration.

References

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